

SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

Fingerprinting Large Language Models (LLMs) is a cornerstone of provenance verification and model attribution. Yet existing fingerprinting methods are evaluated almost exclusively after fine-tuning, once models have already acquired stable signatures from training data, optimization dynamics, or hyperparameters. This is the wrong regime: most of a model’s capacity and knowledge are acquired during pretraining rather than downstream fine-tuning, so large-scale pretraining is the fundamental regime in which lineage must be verified. We demonstrate that existing fingerprinting methods break down in this regime, because they depend on post-hoc signatures that only emerge after substantial training. This failure squarely contradicts the classical Galton notion of a fingerprint as an intrinsic and permanent identity. We therefore introduce a stronger and genuinely intrinsic notion of LLM fingerprinting: **SeedPrints**, a method that turns random initialization biases into persistent, seed-dependent identifiers that are present before a single training step. We show that untrained models already carry reproducible prediction biases fixed by their initialization seed, and that these signals remain statistically detectable throughout training, delivering high-confidence lineage verification. Where prior techniques collapse during early pretraining or degrade under distribution shifts, **SeedPrints** remains effective across every training stage, from initialization through large-scale pretraining to downstream adaptation. Experiments on LLaMA-style and Qwen-style models establish seed-level distinguishability and enable end-to-end birth-to-lifecycle identity verification. Evaluations on large-scale pretraining trajectories and real-world fingerprinting benchmarks confirm its robustness under prolonged training, sharp domain shifts, and parameter modifications.

Together, these results establish that initialization itself imprints a unique and persistent identity on LLMs, constituting a true “Galtonian” fingerprint. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprints have emerged as a key tool to identify, attribute, and trace LLMs by examining their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). Their promise is concrete: to give model owners a verifiable link between a suspicious model and its putative original, and thereby to expose model theft or unauthorized reuse (Yoon et al., 2025; Zhang, 2025).

Much of this literature explicitly borrows the metaphor of biological fingerprints from Francis Galton’s *Finger Prints* (1892) (Galton, 1892):

“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”

The analogy sets a clear bar: an effective fingerprint must be both unique and persistent, present from the very moment of a model’s “birth” at initialization. Measured against this bar, essentially all existing so-called fingerprinting approaches fall short (Pasquini et al., 2024; Xu et al., 2024; Yoon

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

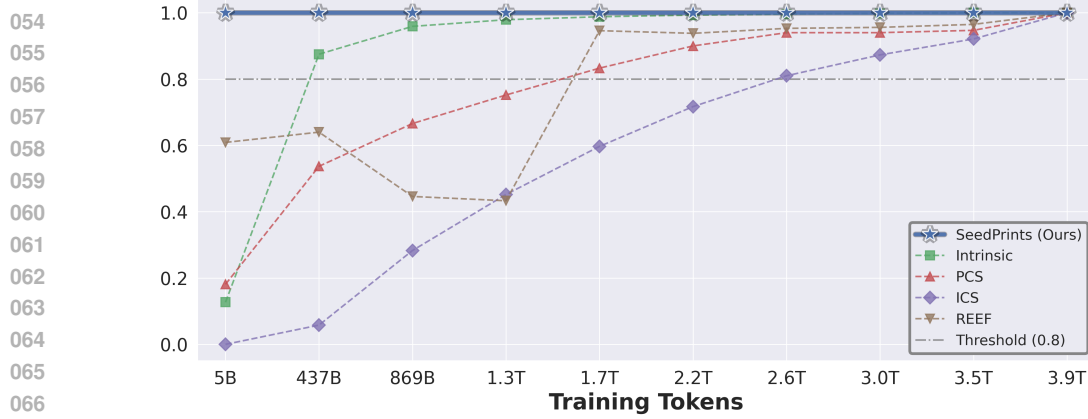


Figure 1: **Existing fingerprinting methods fail to detect model lineage during early pre-training.** We compare five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each tested against the final checkpoint. The y-axis shows the similarity score (higher indicates a stronger lineage signal); the dashed line marks the 0.8 detection threshold. Every baseline degrades and falls below the threshold at early checkpoints, whereas **SeedPrints** achieves perfect detection ($p \ll 0.001$, plotted as $1 - p$) from the very first checkpoint onward.

et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As Figure 1 makes plain, these methods simply cannot detect lineage during early pretraining. The reason is structural: they are defined only after models have fully converged, e.g., by extracting patterns from parameters or generated text. Consequently, the separability they achieve is not a birthmark of the model itself but an imprint of surface-level training components such as data signatures, optimization dynamics, or hyperparameters. Indeed, in Section 5.1, existing baselines break down and misattribute lineage once the training data distribution changes significantly. Such methods are therefore post hoc identifiers, not **Galtonian fingerprints**—the innate, immutable marks that accompany a model from its very beginning.

In this work, we establish a stricter and more principled notion of LLM fingerprinting: an intrinsic property present at *initialization* and detectable at *any* point during subsequent training. Our key contributions are:

- **Exposing a fundamental limitation of existing methods.** We prove empirically that prior fingerprinting approaches cannot reliably detect lineage during early pretraining and are readily misled by substantial shifts in the training data distribution. (Section 5.1)
- **Discovering and harnessing initialization-driven fingerprints.** We uncover that untrained models already exhibit seed-dependent biases in their internal representations—a weak but persistent identity signal fixed at initialization. (Section 3) Building on this, we introduce **SeedPrints**, a method that isolates these initialization-born fingerprints and stays robust to confounding factors such as data distribution, training duration, and optimization dynamics. (Section 4)
- **Extensive empirical validation.** We show that our method separates models differing only in initialization seed, even under identical training pipelines and data orders, and that it persists under continual training across diverse datasets. (Section 5.1)
- **Strong performance in realistic settings.** Our method reliably detects lineage throughout large-scale pretraining trajectories (e.g., OLMo-2-7B Stage 1), with no early-stage failure. It also performs strongly in large-scale fine-tuning settings (e.g., up to 700B tokens) and practical deployment scenarios on the *LeaFBench* benchmark (Shao et al., 2025), which spans 65 models across 7 mainstream model families and 6 transformations (e.g., instruction tuning, fine-tuning, PEFT, quantization, model merging, and distillation). Here our method matches the strongest baseline (near-perfect) while decisively outperforming all others (Sections 5.2 and 5.3).

2 RELATED WORK

LLM protection broadly falls into two families: (i) *watermarking / active fingerprinting* methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* a signature from a

model’s pre-existing behaviors without modifying it (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025).

Watermarking and Active Fingerprinting. Active approaches deliberately implant verifiable identifiers for later ownership checks. Classic techniques include backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018), digital signatures and hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, two common forms are: *text watermarks*, which bias generation or insert predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.); and *model weight watermarks*, which embed identifiers into parameters or link them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints are relatively straightforward and can persist after moderate fine-tuning (e.g., Dasgupta et al. 2024), these invasive schemes fundamentally require control of the training process, which rules them out for retroactively marking third-party models.

Passive LLM Fingerprinting. In contrast, passive fingerprinting identifies models by analyzing their intrinsic properties without any modification. Passive fingerprinting techniques vary by model access. With **white-box access**, signatures are extracted from model weights, leveraging intrinsic properties like the distribution of attention matrices (Yoon et al., 2025), the kernel alignment of internal representations (Zhang et al., 2024), or the stable direction of parameter vectors. In the **black-box setting**, fingerprinting relies on analyzing input-output behavior. These methods use crafted queries (Pasquini et al., 2024), unique prompt-response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025) or iterative prompting–response games (Iourovitski et al., 2024) to identify a model, but remain brittle under fine-tuning. Crucially, all of these methods define their signatures *post hoc*: they read off emergent properties of a completed training process rather than the innate, “Galtonian” fingerprints present from random initialization that our work is the first to isolate. As a direct consequence, *they are simply not designed for lineage verification during large-scale pretraining*, where such post-training signals have not yet emerged.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

In this section, we establish our key observation: language models carry strong prediction biases that originate from the random initialization seed, and these biases remain detectable even after training.

Initialization induces seed-specific bias profiles. We evaluate a LLaMA-2–style model initialized with seed 123 on 10,000 random input sequences of length 1,024, where tokens are sampled uniformly from the vocabulary. In Figure 2 (Left), we plot the frequency with which each output token (upper, from the final logits) and each hidden dimension (lower, from the last-layer representations) attains the minimum value across the 10,000 random trials. The pronounced non-uniform shape in both plots is unambiguous evidence of extreme output preference bias patterns. Repeating the experiment across different initialization seeds reveals a clean structure: the overall magnitude of bias stays stable, but the specific argmin dimensions are determined by the seed (Figure 2, Lower Right). These observations align directly with recent findings that biases in randomly initialized models arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and further amplified by self-attention (Li et al., 2026). Under this view, each initialization seed induces its own contraction direction, producing distinct argmin patterns across output dimensions.

Training preserves initialization-born bias Although training substantially changes output magnitudes, the relative preference over output dimensions fixed at initialization survives. We train the previously initialized model (seed 123) for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate intermediate checkpoints. We focus on the $m = 50$ output dimensions that the initialized model most frequently assigns minimum values. For each checkpoint, we compute the correlation between its responses and those of the initialized model across random inputs, restricted to these dimensions. Intuitively, this measures how similarly the checkpoint and the initialized model rank or disfavor these dimensions across inputs. As a baseline, we compute the same correlation with other independent initialized models (seed 1000). As Figure 2 (Upper Right) shows, even though the absolute correlation values are small, correlations for models from the same initialization are consistently shifted upward relative to the independent baseline and stabilize rather than decaying to zero. This is decisive: initialization leaves a weak but statistically detectable bias signal that persists throughout training.

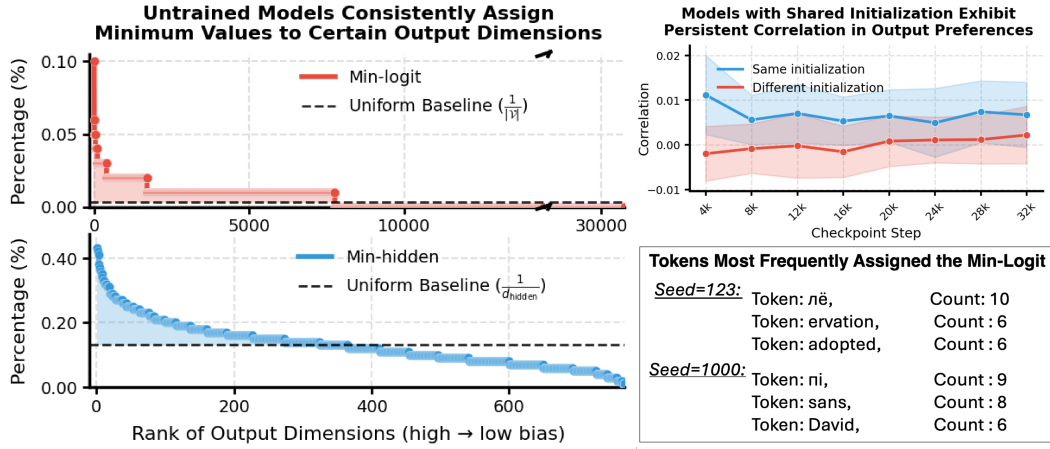


Figure 2: **Initialization-born output bias persists through training.** *Left:* Given completely random inputs, the outputs of a randomly initialized LLaMA-2-style model are far from uniform and instead exhibit pronounced bias: certain dimensions are systematically disfavored by the model (i.e., they frequently receive the minimum value across random inputs). This extreme bias appears both in the logits (top, red) and in the final hidden representations (bottom, blue). The dashed line shows the expected frequency under a uniform distribution. The arrows in the top panel indicate a broken x-axis that omits low-frequency tail ranks. *Upper Right:* Throughout training, models stay weakly but consistently correlated in their output bias across inputs with the base model they are trained from. This correlation distribution is reliably shifted upward relative to correlations computed with independently initialized base models, showing that initialization-born bias leaves a measurable signal that survives training. *Lower Right:* The specific dimensions that exhibit strong bias are determined by the initialization seed; here we illustrate this using tokens that most frequently receive the minimum logit.

4 ALGORITHM

Section 3 establishes that models from the same initialization lineage retain weak but consistent agreement on a subset of output dimensions (e.g., those most disfavored by the untrained model under random inputs). We exploit this directly: our goal is to recover, from any two given models (without access to the true initialized model), the dimensions that encode these bias signals, and to test whether their agreement is statistically significant. The full procedure is summarized in Algorithm 1, and we describe each component below.

Extract identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, where each $x_i \in \mathbb{R}^{\ell \times d}$ denotes a random sequence of length ℓ . Each random input x_i can be instantiated either by uniformly sampling ℓ token IDs from the vocabulary, or by directly sampling ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector as $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality. Here, $g(x_i)$ can denote either the model’s logits or its final-layer representations. We then identify its disfavored dimensions as the set of bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We extract bias signals from the mean response vector because it is a strictly more stable and noise-robust choice than per-sample argmin frequency (i.e., counting how often each dimension attains the minimum value, as illustrated in Figure 2), and it performs well empirically. We make this advantage precise in Appendix C.1.

Given two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. Intuitively, two models sharing the same initialization lineage independently identify a similar set of disfavored dimensions, yielding non-trivial overlap in their bottom- m sets. This intersection-based criterion thus performs mutual verification: it suppresses spurious correlations arising from architectural similarity or shared training dynamics, and isolates precisely the bias signals that originate from shared initialization lineage. When one of the models is closer to initialization, its disfavored dimensions serve directly as an anchor.

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{T}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

Correlation statistics. Restricting both models to the identity dimensions $\mathcal{S}$, we compare their responses across random inputs.¹ For each $j \in \mathcal{S}$, we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n).$$

This yields a set of correlation statistics $T = \{\tau_j\}_{j \in \mathcal{S}}$ that captures the distribution of agreement between the two models across identity dimensions. As established in Section 3, models from the same lineage exhibit distributionally higher agreement on these dimensions than unrelated models do.

Hypothesis testing via null distribution. Because the correlation distributions for same-lineage and different-lineage models can overlap, a simple threshold on individual statistics is inadequate. We instead test whether the observed correlations are significantly larger than expected under the null hypothesis of no lineage relation. Ideally, this null is estimated from the correlation distribution between independent models as in Figure 2. Obtaining such a baseline empirically, however, requires comparisons across many independently initialized models and is computationally expensive. As we show in Appendix B, the null correlation distribution between independently initialized models is well approximated by a Gaussian. This motivates a far cheaper simulation-based alternative: constructing an empirical null distribution using Gaussian surrogates passed through the same pipeline, as summarized in Algorithm 1.

While this simulation-based null already gives a strong empirical approximation, it carries avoidable randomness and computational overhead: its precision is capped by the number of simulation trials, results vary across runs, and comparing two empirical distributions requires additional assumptions from the adopted test, such as the one-sided t-test or the Mann–Whitney U test. Since the null distribution of Kendall–Tau (under weak independence) is available in closed form, we go further and replace the simulation-based procedure with an exact analytical test, while still providing both empirical and analytical implementations in our code repository. Specifically, under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|\mathcal{S}|$ increases, the distribution of $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$ is well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$. Although softmax normalization introduces weak dependencies across dimensions, a relatively high temperature ($T = 10$) suppresses these effects, so the independence assumption holds to good approximation in practice. We empirically confirm that this Gaussian approximation closely matches the distribution obtained from the full pipeline (Appendix B). We therefore compute the one-sided z-score $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$, and the corresponding p-value $p = 1 - \Phi(z)$, where Φ is the standard normal cumulative distribution function. We reject the null hypothesis and declare shared lineage if $p < 0.01$.

Practical instantiation. Model outputs can be taken as either logits or hidden representations. Both are consistent with our formulation, but using logits demands far more random inputs to reliably

¹Before computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This removes scale differences across different models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

Table 1: Different initialization seeds yield clearly distinct fingerprint behaviors. Table 2: Trained models retain the exact fingerprint behavior of their initialization (p -value < 0.01).

Model Pairs	p -value (> 0.01)	Model Pairs	SeedPrints	Baselines			
			p -value	Intrinsic	REEF	PCS	ICS
s_{42} VS. s_{2000}	0.116	s_{42}^{init} vs. s_{42}^{base}	3.37e-26✓	-0.021×	0.375×	0.580×	0.196×
s_{123} VS. s_{42}	1.000	s_{123}^{init} vs. s_{123}^{base}	3.16e-33✓	0.149×	0.369×	0.581×	0.188×
s_{1000} VS. s_{123}	0.327	s_{1000}^{init} vs. s_{1000}^{base}	1.12e-21✓	-0.252×	0.381×	0.581×	0.188×
s_{2000} VS. s_{1000}	1.000	s_{2000}^{init} vs. s_{2000}^{base}	3.70e-31✓	-0.337×	0.331×	0.581×	0.188×

estimate bias patterns as the vocabulary grows, incurring substantial computational overhead (see Appendix C.2 for the theoretical analysis). For efficiency and stability, we therefore adopt final-layer hidden states as the default model output. The resulting p -value delivers a direct statistical test for lineage, with no reliance on heuristic similarity thresholds.

5 EXPERIMENTS

This section has two parts. In Section 5.1, we demonstrate that our method is a genuine fingerprint: (i) it enables *birth verification at the seed level*, and (ii) it stays *verifiable across the entire training lifecycle*. Existing baselines, by contrast, cannot support verification at early pretraining stages and rely predominantly on data-dependent signals that collapse under significant distribution shifts. We report results using the analytical null with *hidden state* outputs, and provide additional results based on the empirical null with both *logit* and *hidden state* outputs, evaluated using the one-sided t -test and Mann–Whitney U test in Appendix D.2. In Section 5.3, we evaluate all methods under practical infringement scenarios using *LeaFBench* (Shao et al., 2025). Across 65 distinct model instances spanning 6 representative post-development techniques, our method matches the best baselines in post-training settings while remaining robust to diverse deployment transformations.

Baselines We mainly consider four passive fingerprinting baselines (weight- or representation-based). *Intrinsic fingerprint* (Yoon et al., 2025) (or *PDF* in some papers) compares models via the similarity of the layerwise standard-deviation profiles of attention parameters. *REEF* (Zhang et al., 2024) computes centered-kernel-alignment (CKA) similarity between feature representations from the same samples across two models. *PCS* and *ICS* (Zeng et al., 2024) (or collectively as *HuRef* in some papers) are weight-similarity methods: PCS flattens all parameters and measures cosine similarity; ICS forms invariant terms from the weights and measures cosine similarity on those invariants. In Section 5.3, we additionally include a (weaker) gradient-based fingerprint *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), we use a **0.8 similarity threshold** for binary decisions.

Note, in all experiment tables, cell colors indicate lineage: with green denotes models from the same source, and red denotes different sources. For example, s_{42}^{init} vs. s_{42}^{base} compares a model initialized with seed 42 and its continued-pretrained counterpart, hence green. By contrast, s_{2000}^{init} vs. s_{42}^{base} compares a seed-2000 initialization with a model trained from a seed-42 initialization, hence red. Additionally, ✓ denotes a correct detection, while × denotes an error.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main paper presents results for LLaMA-style models and defers those for Qwen-style models to Appendix D; the conclusions are identical across the two.

Different initialization seeds produce distinct fingerprints Table 1 reports p -values from our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000). Every p -value is > 0.01 , so our method reliably separates models trained from different seeds. Distinct seeds yield distinct fingerprint behaviors, and models can be told apart “at birth.”

Training preserves the initialization fingerprint. Table 2 compares each initialization model s^{init} with its descendant s^{base} trained on the OpenWebText dataset (Gokaslan et al., 2019) (≈ 10 B tokens). Across all seed–model pairs, p -values are uniformly < 0.01 : their bias profiles remain strongly correlated and therefore share a common lineage. In short, the trained model inherits the exact

Table 3: Even identical data and training order cannot make fingerprint behaviors converge across different initializations.

Model Pair	p -value (> 0.01)
s_{42}^{init} vs. s_{123}^{base}	0.573
s_{123}^{init} vs. s_{2000}^{base}	0.724
s_{2000}^{init} vs. s_{1000}^{base}	0.109
s_{1000}^{init} vs. s_{42}^{base}	0.883

Table 4: The fingerprint persists under continual training on diverse datasets (base model: seed 1000, corpus openwebtext).

Setting	SeedPrints	Baselines			
Continual corpus (seed)	p -value	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	≈ 0 ✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	≈ 0 ✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	1.000✓	0.445✓	0.580✓	0.301✓	0.026✓

fingerprint of its initialization. We also evaluate the baseline methods, and without exception they fail to distinguish across seeds—direct evidence that their separability stems from training-induced artifacts rather than initialization.

Identical data and order do not make fingerprints converge In Table 3, all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus* (OpenWebText) and in the *same data order* (we fix the training seed to lock the data order); the sole difference is their initialization seeds i . Across all cross-seed pairs, p -values stay > 0.01 , in sharp contrast to the near-zero values in Table 2. The conclusion is unambiguous: fingerprints remain seed-specific even under identical data and curriculum.

Continual training on diverse datasets does not confound the fingerprint The experiments above establish that SeedPrints acts as a biometric fingerprint. From a copyright perspective, the inability of prior works to distinguish models with different seeds is not itself a weakness, since initialization seeds carry no clear copyright status. Their real fragility lies elsewhere: their attribution is easily misled by *data distribution, failing to recognize lineage whenever the training distribution shifts substantially during continued training*.

In Table 4, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two very different datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base, versus (ii) *distractors* derived from a different base model (seed 123). Since baseline methods are not sensitive to initialization differences, the distractor base is pretrained with a different data order on OpenWebText, before being continually trained on the *same* target corpus to form the distractor. The question is whether attribution methods can identify which descendant truly shares lineage with the base.

Every prior baseline fails under the code setting (The Stack), misclassifying true descendants as distractors. This exposes what they actually measure—domain similarity, not lineage identity: TinyStories stays close in distribution to the pretraining corpus (OpenWebText), while The Stack diverges sharply, and such a large distribution shift readily bypasses detection. Our method, by contrast, attributes lineage correctly across both corpora. Our fingerprint is therefore no proxy for data distribution: it survives substantial domain shift and persists well beyond the initial pretraining stage.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. Can our method reliably identify its offspring along the training trajectory? It can. We confirm this in a controlled pretraining experiment on OpenWebText, where the initialization seed is known: testing lineage at intermediate checkpoints yields consistent detection at every stage (Appendix D.2.1, Figure 5).

Large-scale pretraining. To evaluate lineage under realistic foundation model training, we analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B $\rightarrow$ 3.9T tokens), treating the final checkpoint as the target model and all previous checkpoints as tested ancestors. Unlike the controlled OpenWebText setting, this stage spans heterogeneous corpora, far longer optimization, and large-scale representation drift. We plot $1 - p$ for consistent directionality with similarity-based baselines. All p -values are $\ll 0.01$ and numerically indistinguishable from zero.

Figure 1 makes the contrast stark: **every existing baseline fails to verify lineage during early pretraining** (e.g., ICS ≈ 0 , PCS < 0.3 and REEF < 0.6 within the first **trillion** tokens), whereas

Table 5: Fingerprinting results under large-scale finetuning. Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the p -value from our correlation test (< 0.01 signals decisive lineage evidence). The four baselines all report similarity scores (threshold = 0.8, higher = better).

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000 ✓	0.9950 ✓	0.9979 ✓	0.9952 ✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000 ✓	0.9985 ✓	0.9985 ✓	0.9949 ✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000 ✓	0.9979 ✓	1.0000 ✓	0.9994 ✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990 ✓	0.9978 ✓	1.0000 ✓	0.9817 ✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480 ✓	0.9947 ✓	0.6863 ✗	0.3369 ✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470 ✗	0.9984 ✓	0.6682 ✗	0.2905 ✗

SeedPrints stays consistently verifiable and only strengthens as training proceeds. This exposes a critical blind spot in existing evaluations: **most prior work verifies lineage only after fine-tuning, precisely where detection is easiest.** Our findings show that large-scale pretraining itself dramatically reshapes representations and inflates lineage signals, granting a dangerous *false sense of safety*. Lineage verification must therefore target the early stages of pretraining—where misuse is hardest to detect—rather than leaning on late-stage checkpoints.

Large-scale finetuning. We further compare our method against existing baselines under standard fine-tuning-stage evaluations. We test suspect models fine-tuned from Llama-2-7b (base model) with data volumes ranging from 5 million to 700 billion tokens. The suspects span diverse downstream variants: Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), Code-Llama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023), with fine-tuning data volumes of 5M, 370M, 1.8B, 13B, 500B and 700B tokens, respectively. As Table 5 shows, our method holds $p < 0.01$ across every setting.

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployments routinely apply *parameter-altering* adaptations to foundation models (e.g., fine-tuning, PEFT, quantization) that can weaken or distort fingerprint signals. Our method is built to remain effective precisely in this regime. To evaluate its reliability under these realistic conditions, we adopt *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing.

Across Source Models. We consider two types of source models for auditing: *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each source type, the task is to distinguish *derivative* models (obtained by post-training adaptations) from *independent* models (trained without using the source weights). This directly mirrors real auditing scenarios where one must test lineage claims for either a PT base or its IT variant. Our evaluation covers **65** models in total; the full list of model names and HuggingFace repositories appears in Appendix E.

Metrics. We evaluate model detection performance using (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), which measures the maximum separation between score distributions of derivative and independent models.

Our method produces per-model p -values from statistical tests, which are fundamentally different from the similarity scores used by baseline methods. Similarity scores are linear measures where differences carry proportional meaning; p -values are tail probabilities of the rarity of an observation under the null hypothesis and cannot be read on a linear scale. To enable comparison with baseline methods that compute AUC from similarity scores, we convert our p -values into scores via $s = 1 - p$. This conversion is deliberately conservative for our method and, if anything, understates it: small p -values change on an exponential scale while threshold sweeping is linear, blurring fine-grained differences. Even under this handicap, our approach matches or exceeds the baselines. For evaluation, we (i) compute AUC for all methods by sweeping thresholds over the scores s , and (ii) report the KS statistic as a threshold-free measure of distributional separability. This protocol guarantees compatibility with baseline pipelines while preserving the statistical interpretability of our method. We do not report Manhattan distance, as absolute distances between p -values are not meaningful.

Note that, in most practical scenarios, calibrating an optimal similarity threshold is simply infeasible. While we report AUC for comparability, our method differs from prior methods in a decisive way: it

Table 6: Performance comparison of LLM fingerprinting methods across different source models. “PT” and “IT” refer to using the pre-trained models and instruction-tuned models as source models, respectively. The per-family breakdown covers the 6 families that have both PT and IT models. The “Overall” scores are computed over all test model pairs, not averaged across families.

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC $\uparrow$	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic $\uparrow$	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC $\uparrow$	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic $\uparrow$	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC $\uparrow$	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC $\uparrow$	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic $\uparrow$	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

provides a direct statistical test. **Given any pair of models, our approach outputs a p -value that yields a definitive decision on whether they share the same lineage, with no tunable thresholds, guaranteeing reliable verification in practice.**

Parameter-Altering Techniques. To assess fingerprinting robustness, we evaluate models produced via (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

5.3.1 FAMILY-WISE EFFECTIVENESS

There are seven model families in total. For each family, we treat both its pretrained base model (PT) and its instruction-tuned model (IT) as separate source models and compare them against all other models in the pool of 65. Table 6 reports the results. We omit the TinyLlama-1.1B families in Table 6, as they are trivial cases for all methods. Each column corresponds to one family (AUC is computed over the corresponding tests with a family-calibrated threshold), whereas the overall score is computed across all test pairs with a global threshold; the overall score is therefore not the average of the per-family scores. Three methods—*Intrinsic*, *ICS*, and our *SeedPrints*—are saturated across all alterations (overall AUC of 0.994, 0.994, and 0.992, respectively; KS of 0.989, 0.943, and 0.986), with family-wise scores at the full score in most cases. The remaining two baselines, *REEF* and *Gradient*, trail substantially. The takeaway is clear: although our method is designed to capture intrinsic seed-level fingerprints, it also functions as a biometric-grade fingerprint, delivering reliable and persistent identity tracking across diverse model families.

5.3.2 ALTERATION-WISE ROBUSTNESS

Across alteration types, *Instruct* is consistently the easiest case (even the weakest baseline, *Gradient*, reaches AUC 0.895), whereas *Finetune* and *Merge* are the hardest for all methods—no method attains perfect *Finetune* performance. The effects of *PEFT*, *Quantization*, and *Distillation* are more mixed: REEF drops sharply under *Quantization* (0.996 $\rightarrow$ 0.871) and *Distillation* (0.996 $\rightarrow$ 0.858) while staying resilient to *PEFT* (0.994); *PEFT* and *Quantization*, conversely, are harder for *Gradient* (0.895 $\rightarrow$ 0.776/0.756). The three stronger methods stay robust across every transformation (all scores $>$ 0.9). *SeedPrints* dips only to 0.959 on *Merge*, missing a single merge model out of seventeen; this lone misclassification indicates that weight-space interpolation with unrelated model weights can dilute the initialization signal below the detection threshold. Averaged across all six alteration types, *Intrinsic* attains the highest overall AUC, with *ICS* and *SeedPrints* following closely behind.

5.4 ABLATION STUDY ON HYPERPARAMETERS

Our method has three hyperparameters: the number of random input sequences n , the length of each random sequence l , and the fingerprint size m . The first two control the overall size of random inputs, while m sets how many low-mean coordinates are used for testing. Throughout the main experiments in Sections 5.2 and 5.3, we fix $n=2000$, $l=1024$, and $m=0.1d_{\text{out}}$, where d_{out} denotes the hidden state dimensionality, which lets the method adapt cleanly to different open-source models.

Table 7: Performance of LLM fingerprinting methods across a range of parameter-altering techniques.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC $\uparrow$	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic $\uparrow$	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC $\uparrow$	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic $\uparrow$	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC $\uparrow$	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic $\uparrow$	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC $\uparrow$	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic $\uparrow$	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC $\uparrow$	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic $\uparrow$	1.000	0.990	0.980	1.000	0.941	1.000	0.985

Here we vary each hyperparameter while holding the others at their default values (i.e., $n = 2000$, $m = 400$, $l = 768$), and report empirical accuracy and false positive rates under a fixed significance level $\alpha = 0.01$ in Table 8.

Increasing n and l consistently improves identification performance. Our theoretical analysis in Section C.2 explains exactly why: accurate recovery of the underlying population-level bias requires $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$, where γ is the bias margin separating biased coordinates from the remaining dimensions. A target output dimension thus needs a minimal number of samples for accurate detection, which is precisely why small n underperforms. Moreover, γ grows with l through accumulation effects, so longer sequences sharpen the separation. Consequently, under a fixed (insufficient) number of sequences, increasing l lowers the required query count and improves detection.

The effect of m is non-monotonic. Too small an m leaves the selected coordinates unstable, with too few empirical observations to form a reliable distribution. Too large an m pulls in many irrelevant coordinates whose noise shrinks γ , which then demands quadratically more samples to offset. Empirically, m in the range 200–400 works well across model sizes, matching our heuristic of $m \approx 0.1d_{\text{out}}$ for typical $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

In this work, we introduced *SeedPrints*, a stronger, intrinsic notion of LLM fingerprinting that traces a model’s lineage all the way back to its random initialization. We established that initialization itself stamps a persistent “Galtonian” fingerprint on neural language models: untrained models already carry stable, seed-dependent token-preference patterns, and these biases persist—statistically and measurably—throughout training. Building on this, we proposed a distribution-correlation test over *identity dimensions* that detects shared lineage with calibrated significance. Across LLaMA- and Qwen-style families and a broad suite of realistic deployments (e.g., continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation), SeedPrints delivers “birth-to-lifecycle” verification and stays robust under domain shift, providing a practical tool for provenance and copyright auditing.

Limitations and Future Directions This work uncovers fingerprint signals rooted in initialization-stage bias. The proposed method requires access to model outputs (e.g., logits or hidden states) and does not directly apply when only restricted API access is available. We stress, however, that the pursuit of *true fingerprints* for LLMs remains at an early stage: most existing approaches (especially black-box methods) rely on signals that emerge only in later training stages or from alignment-driven artifacts, and thus cannot robustly protect models at the pretrained stage. This makes reliable all-stage fingerprints all the more urgent. See full discussions in Section A.2.

Table 8: Ablation study under significance level $\alpha = 0.01$.

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

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A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We use AI assistants, i.e., ChatGPT and Gemini, for writing and formatting support. Their use covers grammar and style checks, improving clarity of figure and table captions, and other surface-level edits. For programming-related tasks, we occasionally use GitHub Copilot and Claude as coding assistants, e.g., for code auto-completion and debugging hints.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

This work uncovers seed-born, training-persistent biases that serve as robust fingerprint signals, but our method operates within the white-box paradigm. It does not directly apply when only restricted API access is available, for instance when responses are deterministic or safety-filtered. We emphasize, however, that the pursuit of *true fingerprints* for LLMs remains at a very early stage even under the white-box setting: most existing fingerprinting approaches rely on training-induced signatures or alignment behaviors and therefore cannot robustly protect pretrained or early-stage models. By analogy to human fingerprints, which remain stable regardless of age or environment, a true LLM fingerprint must persist across continued training and deployment rather than reflect incidental training artifacts—and SeedPrints is, to our knowledge, the first method to meet this bar across the full training lifecycle. Extending it to black-box access, where a substantial performance gap to white-box methods persists, is a demanding open direction and calls for significant future work. We expect this work to catalyze further research toward principled, robust fingerprinting frameworks for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

Our hypothesis test in Section 4 evaluates whether two models share lineage by comparing the distribution of their correlation statistics over random inputs against a null distribution. The null distribution models the correlation statistics between two *independent models evaluated on random inputs*.

We consider two approaches for approximating this null distribution: (i) an empirical simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. In this section, we justify these approximations through three steps: (i) characterizing the true null using independently initialized models, (ii) introducing the Gaussian surrogate as a tractable simulation-based proxy, and (iii) validating the analytical approximation. Each step confirms that our approximations track the true null closely.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null distribution is the correlation statistics between two independent models evaluated on random inputs. The cleanest way to instantiate this independence is to compare two independently initialized models (with different random seeds and no training). We estimate this empirical null by evaluating ~ 2500 such pairs and computing Kendall–Tau correlations on 10,000 random inputs (as in Section 3). Figure 3 confirms that the true null distribution is empirically a Gaussian centered at zero.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Directly estimating the null distribution using independently initialized models is computationally expensive, as it requires access to many models and repeated evaluations. To obtain a tractable alternative, we introduce a simulation-based null by using Gaussian surrogates as proxies for model outputs under the null.

Two considerations justify this choice. First, approximating neural network outputs with Gaussian distributions is standard practice in the auditing literature, backed by both empirical and theoretical studies (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline depends only on relative ordering

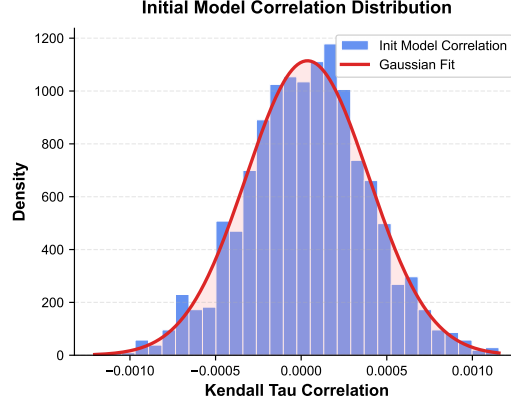


Figure 3: Empirical correlation distribution over ~ 2500 pairs of independently initialized models, evaluated on random inputs, closely matches a Gaussian distribution centered at zero.

and selection operations (e.g., rank correlations and bottom- m selection) and does not rely on any semantic structure in the inputs. Furthermore, randomly initialized models have independent parameter matrices, so their outputs under the null carry no strong structured dependencies. Sampling independent Gaussian matrices therefore provides a simple, tractable, and faithful proxy for the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

This simulation-based construction applies the same selection procedure to Gaussian samples and naturally captures the dependency patterns introduced by the pipeline itself (e.g., those arising from selecting $\mathcal{S}$ across dimensions). As Figure 4 (middle) shows, the resulting distribution matches the empirical null from independently initialized models in its centered Gaussian shape, confirming that the Gaussian surrogate is an effective simulation-based approximation of the true null.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

While the Gaussian surrogate captures the behavior of the null distribution under the full pipeline, it still requires repeated sampling and injects randomness into the estimation process. We therefore also derive an even more efficient analytical approximation.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization introduces dependencies across dimensions. Yet Figure 4 shows the analytical Gaussian closely matches the distribution from the Gaussian surrogate pipeline across every setting, including varying numbers of random inputs n and output dimensions D . These dependencies are therefore weak enough to leave the accuracy of the approximation intact. The likely reason is that the relatively high temperature ($T = 10$) smooths the output distribution and decouples the dimensions.

²As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

Together, these results firmly justify using the analytical Gaussian null in Section 4 as an efficient, deterministic, and accurate approximation that eliminates the need for costly empirical baseline construction.

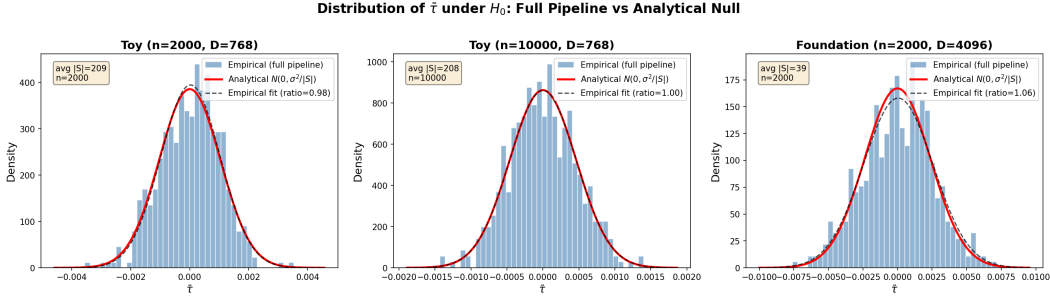


Figure 4: Validation of the analytical Gaussian null. We apply the full pipeline (including identity-dimension selection and correlation aggregation) to Gaussian surrogate outputs and compare the resulting distribution of $\bar{\tau}$ with the analytical Gaussian $\mathcal{N}(0, \sigma^2/|S|)$. The close match confirms that the analytical approximation accurately captures the null behavior, despite the weak dependencies introduced by the pipeline.

On variance scaling. The true null from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This mismatch leaves our hypothesis test results untouched. Our primary evaluation for the simulation-based null in Section D.2 uses the Mann–Whitney U-test, which is rank-based and thus invariant to any positive scaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$. All rank statistics and rejection thresholds are therefore unchanged. For tests that do depend on absolute variance, we additionally evaluate variance-sensitive statistics such as the t -test, whose statistic scales as $t' = t/c$ when variance decreases. Since $c < 1$ in our case, scaling shrinks the magnitude of t and makes the test strictly more conservative rather than inflating significance. Empirically, across the extensive experiments in Section D.2, variance scaling does not diminish the power of the t -test at all—clear evidence that the distributional separation between the alternative and the null far exceeds the scaled variance.

For the analytical null used in the main paper, the variance enters explicitly through the normalization of the z -score, so any scale discrepancy is absorbed in the denominator. Because the analytical variance is slightly larger than the empirical one, the resulting z -scores are smaller, keeping the test conservative and firmly guarding against false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

Our goal is to identify the output dimensions that the initialized model is least likely to predict under random inputs. These dimensions are precisely the directions the model inherently disfavors due to initialization-induced bias (independent of the input distribution), and they form the basis of a stable fingerprint signal.

Setup. Consider a model g with fixed parameters and random inputs x_i (uniform random tokens or random embeddings as in the main paper). For each output dimension $j \in [d_{\text{out}}]$, $g_j(x)$ is thus a random variable and its population mean is $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we aim to capture correspond to the m output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap measures how well-separated the target set $\mathcal{M}$ is from the remaining coordinates. Importantly, for a fixed m and input distribution, $\Gamma(m)$ depends solely on the model’s inherent initialization bias. *However,*

increasing the length of random input sequences amplifies this bias via concentration, making the separation more pronounced at the population level.

There are two natural empirical strategies to extract such population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs $X = \{x_i\}_{i=1}^n$ and select the m coordinates with the smallest value. Following the notation in Section 4, where g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, select the smallest-value coordinate and then take the m most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both approaches look reasonable at first glance. As we prove below, however, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and flips under arbitrarily small perturbations, making it inherently noise-sensitive. This settles the design choice decisively in favor of the mean-based estimator.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first prove that the sample-mean-based operator, which selects the m smallest coordinates of the empirical mean vector, is stable under small perturbations to model outputs. These perturbations represent realistic variations in outputs from the *same* model, e.g., numerical noise or evaluation under different precision modes.

Formally, we model perturbed outputs as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ represents stochastic deviations. We assume that for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\widehat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma gives a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. Crucially, this instability probability decays exponentially in the number of averaged samples n , driven by variance reduction from σ^2 per sample to σ^2/n in the empirical mean.

³This sub-Gaussian assumption is mild: bounded random variables are σ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is a special case of this assumption.

Proof. We condition on the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\hat{\Gamma}(m)$ are fixed, and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ only.

For any coordinate j , by definition of $\hat{g}'$, $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, so their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Hence, for any $\varepsilon > 0$, by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \right\}.$$

Equivalently, if $\max_j |\hat{g}'_j - \hat{g}_j| < \hat{\Gamma}(m)/2$, then the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ such that $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\hat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \hat{\Gamma}(m)$. With $|\hat{g}'_i - \hat{g}_i| < \hat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \hat{\Gamma}(m)/2$, we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\hat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\hat{\Gamma}(m)}{2}) \geq \hat{\Gamma}(m) - \hat{\Gamma}(m) = 0.$$

Thus no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$ under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since $\hat{\Gamma}(m)$ is fixed under conditioning on X , plugging $\varepsilon = \hat{\Gamma}(m)/2$ into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\hat{\Gamma}(m)^2}{8\sigma^2}\right). \quad \square$$

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

By contrast, per-sample voting applies a discontinuous winner-take-all operator to each input. As a result, arbitrarily small perturbations to outputs flip decisions with non-vanishing probability. We again consider the same stochastic perturbation model $g'(x) = g(x) + \eta(x)$.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. Unlike mean extraction, voting reaps no benefit from variance reduction: each input forces a hard decision on noisy outputs. Increasing n merely multiplies the opportunities for index-flips.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for brevity. By definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ be an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, certainly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , because the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$, and $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of the dimensions $\{j^*(x_i; g)\}_{i=1}^n$. If the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions is of order $O(1/n)$ (i.e., only a constant number of votes), then a change in $j^*(x_i)$ for a single input x_i is sufficient to swap the ordering across this boundary and thus alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ shows that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n , i.e., per-sample argmin voting does not enjoy variance reduction with more samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We now analyze how many random inputs are required for the empirical mean estimator to recover the true bias. We reuse notation from **Setup**: each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$, with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ represents the sorted means in non-decreasing order, and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ is the separation margin. We estimate $\bar{g}$ using n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimensions set is $\mathcal{M}$ with size m and our empirical estimation gives $\widehat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\widehat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- m set. $\square$

We next translate this recovery condition into hyperparameters. We again adopt the same sub-Gaussian assumption as in Section C.1. This assumption is mild: by Hoeffding's inequality, any bounded random variable is σ -sub-Gaussian with σ proportional to its range. In our setting, each $g_j(x)$ is a logit or final hidden activation, which is bounded or light-tailed in both initialized networks (due to controlled parameter initialization) and trained networks.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Table 9: Fingerprint behavior is distinctly seed-specific under different random initializations across model families.

(a) LLaMA-style models					(b) Qwen-style models				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: (LLaMA-style models) Trained models retain the exact fingerprint behavior of their initialization (p -value < 0.01).

Model Pair	Logits Output		Hidden State		Baselines			
	<i>t</i> -test	<i>u</i> -test	<i>t</i> -test	<i>u</i> -test	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.33e-3✓	1.02e-3✓	2.20e-8✓	6.28e-8✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	2.06e-3✓	7.33e-3✓	7.09e-6✓	1.37e-5✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	2.44e-3✓	4.14e-3✓	5.58e-4✓	2.81e-3✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	5.63e-3✓	6.76e-3✓	4.00e-10✓	1.27e-9✓	-0.337×	0.331×	0.581×	0.188×

Accurate recovery therefore requires $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$: the number of random queries grows only logarithmically with the output dimension, but scales inversely with the squared separation margin γ . The margin γ itself has two contributing factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias via concentration, increasing the population separation between coordinates.
- **Choice of m .** Different choices of m correspond to different parts of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$, thus yielding different effective margins in practice.

The former can be controlled by the input length l , while the latter is selected empirically based on the observed gaps in the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models with the Hugging Face Transformers Trainer (Wolf et al., 2020), using Accelerate (Gugger et al., 2022) for distributed runs. All open-source models are loaded from their official Hugging Face releases and used under their original licenses: Llama models under the Meta Llama Community License, and other models under Apache-2.0. All datasets are downloaded via the Hugging Face Datasets library (itself Apache-2.0); dataset content follows each dataset’s stated license.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section provides the complementary results for Section 5. We report results using the empirical simulation-based null in Algorithm 1 with both *logits* and *hidden state* outputs, testing with both the one-sided *t*-test (*t*-test) and the Mann–Whitney *U* test (*U* test). Across all of these variants, the conclusions of the main paper hold without exception.

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based random baseline

Table 11: (Qwen-style models) Trained models retain the exact fingerprint behavior of their initialization models (p -value < 0.01).

Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{123}^{base}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s_{1000}^{init} vs. s_{1000}^{base}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s_{42}^{init} vs. s_{42}^{base}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s_{2000}^{init} vs. s_{2000}^{base}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: The fingerprint persists under continual training on diverse datasets (base model: seed 1000, corpus openwebtext). U -test refers to the Mann–Whitney U test.

(a) LLaMA-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0 [✓]	0 [✓]	0 [✓]	7.77e-89 [✓]	1.000 [✓]	0.759 [×]	0.999 [✓]	0.996 [✓]
TinyStories (123)	1.000 [✓]	1.00 [✓]	0.943 [✓]	0.902 [✓]	0.950 [×]	0.658 [✓]	0.332 [✓]	0.012 [✓]
the_stack (1000)	0 [✓]	1.73e-287 [✓]	0 [✓]	3.09e-69 [✓]	0.489 [×]	0.557 [×]	0.585 [×]	0.123 [×]
the_stack (123)	0.616 [✓]	0.479 [✓]	0.732 [✓]	0.831 [✓]	0.445 [✓]	0.580 [✓]	0.301 [✓]	0.026 [✓]

(b) Qwen-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09 [✓]	1.92e-07 [✓]	8.49e-214 [✓]	5.09e-71 [✓]	1.000 [✓]	0.957 [✓]	0.999 [✓]	0.996 [✓]
TinyStories (123)	0.434 [✓]	0.433 [✓]	0.256 [✓]	0.065 [✓]	0.913 [×]	0.199 [✓]	0.328 [✓]	0.039 [✓]
the_stack (1000)	0 [✓]	4.16e-237 [✓]	1.16e-211 [✓]	2.30e-76 [✓]	0.999 [✓]	0.313 [×]	0.995 [✓]	0.976 [✓]
the_stack (123)	0.999 [✓]	0.993 [✓]	0.610 [✓]	0.491 [✓]	0.916 [×]	0.255 [✓]	0.328 [✓]	0.038 [✓]

is stochastic, we report p -values averaged over 10 independent trials and adopt a significance level of $\alpha = 0.01$. Note that the absolute magnitude of extremely small p -values is not meaningful: once p falls below numerical and sampling noise (e.g., $< 10^{-20}$), values like 10^{-260} carry no more evidential weight than 10^{-20} —both decisively reject the null.

Different initialization seeds produce distinct fingerprints Table 9a and Table 9b report p -values from our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under the t -test and U -test, for Llama-style and Qwen-style models respectively. Every p -value is > 0.01 , so our method reliably separates models trained from different seeds. Distinct seeds yield distinct fingerprint behaviors, and models can be told apart “at birth.”

Training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) compare each initialization model s^{init} with its descendant s^{base} trained on the OpenWebText dataset (Gokaslan et al., 2019) (≈ 10 B tokens). Across all seed–model pairs, p -values are uniformly < 0.01 : their bias profiles remain strongly correlated and therefore share a common lineage. In short, the trained model inherits the exact fingerprint of its initialization; training does not erase it. We also evaluate the baseline methods in Table 10, and without exception they fail to distinguish across seeds—direct evidence that their separability stems from training-induced artifacts rather than initialization.

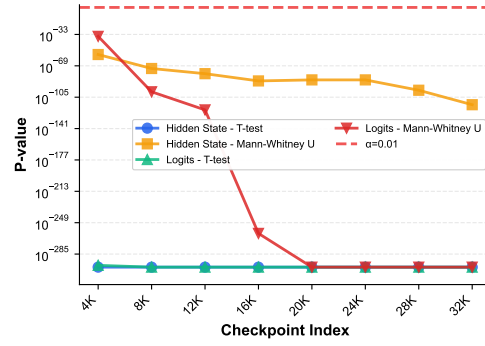
Identical data and order do not make fingerprints converge In Table 13a (LLaMA-style models) and Table 13b (Qwen-style models), all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus (OpenWebText) and in the same data order* (we fix the training seed to lock the data order); the sole difference is their initialization seeds i . This directly tests whether identical data and order can erase or confound fingerprint behavior. Across all cross-seed

Table 13: Even identical data and training order cannot make fingerprint behaviors converge across different initializations.

(a) LLaMA-style models					(b) Qwen-style models				
Model Pair	Logits Output		Hidden State		Model Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486	s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096	s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337	s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287	s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: The fingerprint verifies lineage at every single checkpoint (p -values < 0.01) across model families.

pairs, p -values stay > 0.01 , in sharp contrast to the near-zero values in Table 10. Fingerprints remain seed-specific even under identical data and curriculum.

Continual training on diverse datasets does not confound the fingerprint The experiments above establish that SeedPrints acts as a biometric fingerprint. From a copyright perspective, the inability of prior works to distinguish models with different seeds is not itself a weakness, since initialization seeds carry no clear copyright status. Their real fragility lies elsewhere: their attribution is easily misled by *data distribution*, *failing to recognize lineage whenever the training distribution shifts substantially during continued training*.

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two very different datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (initialized with seed 123 and trained with a different data order on OpenWebText), then continued on the *same* corpus. The question is whether attribution methods can identify which descendant truly shares lineage with the base.

Every prior baseline fails under the code setting (The Stack), misclassifying true descendants as distractors. This exposes what they actually measure—domain similarity, not lineage identity: TinyStories stays close in distribution to the pretraining corpus (OpenWebText), while The Stack diverges sharply, and such a large distribution shift readily bypasses detection. Our method, by contrast, attributes lineage correctly across both corpora. Our fingerprint is therefore no proxy for data distribution: it survives substantial domain shift and persists well beyond the initial pretraining stage.

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We first conduct a controlled pretraining experiment on OpenWebText, where the initialization seed is known. Treating each intermediate checkpoint as the base, we test whether our method can reliably identify its offspring along the same training trajectory. All

Table 14: Cross-size similarity between Qwen-2.5-14B (target) and other Qwen-2.5 models.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

variants recognize the suspect model as same-lineage at every point, with p -values remaining below the 0.01 threshold (Figure 5a for LLaMA-style models and Figure 5b for Qwen-style models). This establishes the stability of SeedPrints under continuous optimization.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To test whether SeedPrints “collapses” within a model family when we vary the parameter size, we report cross-size similarity between Qwen-2.5-14B (target) and smaller models from the same Qwen-2.5 family. Table 14 shows the results for SeedPrint ($\uparrow$) and two baselines, REEF ($\downarrow$) and HUREF ($\downarrow$).

To probe whether initialization-born fingerprints “collapse” into a family-level signature when scaling model size, we conduct a cross-size study within the Qwen-2.5 series. We treat Qwen-2.5-14B as the protected target model and consider Qwen-2.5- $\{0.5B, 1.5B, 3B, 7B\}$ as suspect models that share the same architectural family but are not trained as descendants of Qwen-2.5-14B. As explained in the main paper, SeedPrints outputs a p -value for the hypothesis test “the suspect shares initialization-born fingerprints with the target”, whereas REEF and HUREF are similarity-based baselines that emit continuous scores and force the user to pick a threshold.

Several observations stand out. First, every SeedPrints p -value sits comfortably above a typical significance level (e.g., $\alpha=0.01$). Under our hypothesis-testing view, none of the smaller Qwen-2.5 models is flagged as sharing the same lineage as Qwen-2.5-14B. This is exactly the desired behavior: even though these models belong to the same family and reuse a similar architecture, the fingerprinting test does not collapse to a single family-level signature that would spuriously conflate distinct pre-training runs.

The cross-size p -values spread across different regions of the $[0, 1]$ range rather than concentrating around a single value. We do *not* read these magnitudes as a continuous “similarity score”—their role is to support a calibrated yes/no decision at a fixed significance level. The spread nonetheless demonstrates that our test statistic remains sensitive to the concrete combination of architecture and initialization seed, rather than degenerating into a generic pattern shared by all Qwen-2.5 variants.

Finally, the similarity-based baselines behave very differently. REEF produces scores in a narrow band (0.8250–0.8833) across all sizes, while HUREF outputs values numerically close to zero (on the order of 10^{-5}). Because these methods offer no built-in decision rule, operators must hand-tune thresholds, and no single cutoff robustly separates cross-size variants from genuinely lineage-related models. SeedPrints, by contrast, yields a statistical decision at a prescribed significance level, giving a clear and operational answer to the question “do these two models plausibly share a training lineage?” within a model family.

E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers 58 models in total. Below we document the decoding configuration used throughout all runs and enumerate the model catalog grouped by family. For each derived model we indicate the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*). “PT” and “IT” denote *pre-trained* and *instruction-tuned* source models, respectively.

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Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunnycore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	Xiaojian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2b-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissglc/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” denotes the post-training transformation relative to the PT/IT source model: *finetune* includes supervised/preference optimization variants; *adapter* includes LoRA/QLoRA-style modules; *merge* includes model soups and TIES-style merges; *quantization* includes GPTQ/INTx variants; *distillation* includes student models distilled from reasoning-augmented teachers.

1458 E.2 SAMPLE RESULTS

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1460 We present partial results from one LeafBench evaluation run in Table 15 (u -test with simulation-
1461 based null). Reported metrics are $1 - p$, used as similarity scores. We adopt a significance level of
1462 0.01 for p -values; empirically, LeafBench identifies an optimal decision threshold of 0.9920, which
1463 further confirms the reliability of our method.

1464 Crucially, although individual p -values are not linear similarity measures, their distribution across
1465 model pairs is highly informative: same-lineage models yield much lower p -values, while unrelated
1466 models stay near uniform.

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Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample results.** Pairwise similarity between target models (rows) and base models (columns). Cells with $1 - p \geq 0.99$ are highlighted in green as same-lineage pairs.